

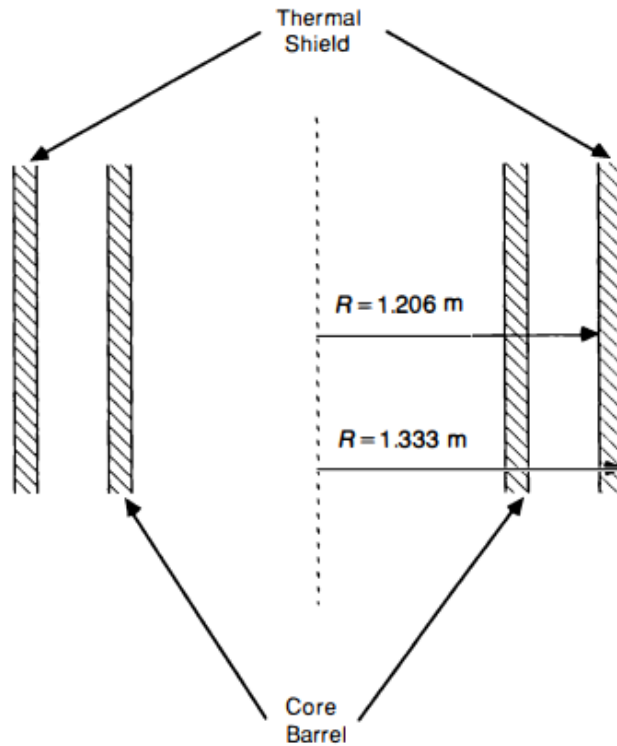


Figure 3-7 Thermal shield.

Example 3-4 Power deposition in a thermal shield

PROBLEM In a PWR the core is surrounded by a thermal shield (Fig. 3-7) to protect the pressure vessel from γ -ray heating and neutron-induced radiation damage. For an iron thermal shield with the radiation values given below, calculate the volumetric heat generation rate in the shield at its outermost position.

Assume that the core of the reactor is equivalent to an infinite plane source and that the shield can be treated as a slab owing to the small thickness-to-radius ratio.

Note that at energies well above the inelastic scattering threshold the total cross section is due to scattering. For steel, the inelastic scattering rate may be assumed to be equal to the elastic scattering rate:

$$\Sigma_{s,e\ell} \approx \Sigma_{s,i\ell} \quad (3-61)$$

The total neutron cross section can be approximated by the sum of both scatterings:

$$\Sigma_T \approx \Sigma_{s,e\ell} + \Sigma_{s,i\ell} \quad (3-62)$$

Use the following values for the radiation flux parameters.